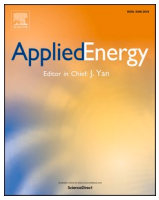
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Toward precision and intelligence: The convergence and evolution of artificial intelligence, data science, and control theory in building HVAC load forecasting

Dapeng Chen [a], Yanfeng Gong [b,*], Jianan Chen [c], Rui He [d]

[a] College of Civil Engineering, Nanjing Tech University, Nanjing 211816, China
[b] College of Urban Construction, Nanjing Tech University, Nanjing 211816, China
[c] College of Electrical Engineering and Control Science, Nanjing Tech University, Nanjing 211816, China
[d] Jiangsu Province Architectural D&R Institute Co. Ltd., Jiangsu 210019, China

HIGHLIGHTS

- Review of 151 studies shows HVAC prediction improved from 15% to 2–5% error.
- Data preprocessing quality determines model performance more than algorithm choice.
- Hybrid deep learning with signal decomposition drives best accuracy improvements.
- Prediction-control integration achieves 10–30% energy savings with comfort gains.
- Physics-informed AI and federated learning address key generalization challenges.

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ABSTRACT

Accurate prediction of heating, ventilation, and air conditioning (HVAC) load demands has become a cornerstone technology for achieving building energy efficiency, enhancing power grid stability, and advancing carbon neutrality objectives in the built environment. This comprehensive review systematically examines the transformative evolution of HVAC load prediction methodologies over the past decade, with particular emphasis on the convergence of data science, artificial intelligence (AI) algorithms, and modern control theory. Through critical analysis of 96 peer-reviewed publications, this study traces the progression from traditional statistical approaches to sophisticated hybrid frameworks integrating advanced data preprocessing, machine learning algorithms, and intelligent control strategies. The review reveals significant accuracy improvements, with Mean Absolute Percentage Errors declining from over 15% in early traditional methods to approximately 2–5% under real-world conditions and below 1% in simulation-based environments, although cross-study comparisons require caution due to heterogeneous datasets and validation protocols. Key findings demonstrate that data preprocessing and feature engineering serve as critical enabling foundations whose quality often determines model performance potential, while the systematic integration of signal decomposition with hybrid deep learning architectures constitutes the primary driver of accuracy improvements. The integration of prediction capabilities with Model Predictive Control and reinforcement learning has demonstrated 10–30% energy savings with simultaneous comfort improvements. Persistent challenges including generalization limitations, uncertainty quantification deficits, and interpretability concerns are identified, alongside emerging opportunities in physics-informed AI, Large Language Model integration, and federated learning, all at early maturity stages for HVAC applications. This review uniquely contributes by: (I) establishing data preprocessing as a critical enabling foundation through systematic examination of signal decomposition and feature engineering; (II) tracing the complete algorithmic evolution from traditional methods through hybrid deep learning frameworks; (III) examining prediction-control integration with Model Predictive Control and deep reinforcement learning; and (IV) critically assessing emerging paradigms with explicit maturity evaluations. The future of building energy management lies in autonomous, adaptive systems combining physics-based understanding with AI adaptability,

* Corresponding author.
E-mail address: gyfnjtech@163.com (Y. Gong).

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optimized control with uncertainty-aware decision-making, and individual building intelligence with community-scale coordination.

1. Introduction

1.1. Global energy consumption and the building sector challenge

To ensure clarity and consistency throughout this review, it is important to define key terminology at the outset. The term "HVAC load" in this review refers primarily to the thermal load, i.e., the cooling or heating energy demand (in kW or kWh) that the HVAC system must deliver to maintain desired indoor environmental conditions. This is distinct from the electrical load consumed by HVAC equipment (compressors, fans, pumps), which depends on the thermal load as well as equipment efficiency (Coefficient of Performance, COP) and operating conditions. Where reviewed studies address electrical energy consumption rather than thermal demand, this is explicitly noted. Furthermore, "building load" encompasses total building energy consumption including HVAC, lighting, plug loads, and other end uses, and is distinguished from "HVAC load" throughout this review. The primary focus of this review is on HVAC thermal load forecasting, though relevant studies on building-level energy prediction and HVAC electrical consumption are included where they contribute to the methodological discussion.

Sustained growth in global energy demand, coupled with mounting environmental concerns and the urgent imperative for carbon neutrality, has positioned the building sector at the forefront of sustainability discourse [25,36]. Buildings are responsible for approximately 40% of global energy consumption and contribute significantly to greenhouse gas emissions, with commercial and residential structures representing one of the most promising yet challenging domains for energy optimization interventions [24,36]. Within this context, heating, ventilation, and air conditioning systems emerge as the predominant energy consumers, typically accounting for 40–60% of a building's total energy consumption and over 25% of peak electricity demand in major metropolitan areas [2,24].

The challenge becomes particularly acute when considering the rapid urbanization trends and the corresponding proliferation of climate-controlled built environments worldwide. Contemporary building stock continues to expand rapidly, while existing infrastructure often operates with suboptimal efficiency due to outdated control strategies, inadequate system integration, and the persistent mismatch between designed capacity and actual operational requirements [2,4]. This operational inefficiency is compounded by the inherent complexity of building energy systems. These systems involve multivariable interactions, nonlinear dynamics, and significant temporal variability driven by occupant behavior, weather fluctuations, and equipment degradation [1,6].

1.2. HVAC system energy consumption: Current status and underlying causes

Modern HVAC systems present a complex web of interconnected challenges that demand sophisticated analytical approaches for resolution. Current operational paradigms in building climate control are predominantly reactive, relying on conventional feedback control strategies that respond to deviations from predetermined setpoints rather than anticipating and proactively managing thermal loads [19,20]. Such a reactive approach inherently introduces control lag, leading to energy waste during transient periods and suboptimal system performance during peak demand scenarios [48,60].

Real-world HVAC systems operate under partial-load conditions for more than 85% of their lifetime, yet their control strategies are typically optimized for full-load scenarios [2]. This mismatch between design assumptions and operational reality creates inefficiencies that accumulate over the building's lifecycle. Moreover, the increasing integration of renewable energy sources, distributed generation, and smart grid technologies has introduced additional layers of complexity that traditional control methodologies struggle to accommodate effectively [34,42].

Economically, implications of these inefficiencies are substantial, with studies indicating that optimized predictive control strategies can reduce HVAC energy consumption by 10–30% compared to conventional approaches [4,7,20,54]. Beyond direct energy savings, improved load prediction and control enable buildings to participate in demand response programs, peak shaving initiatives, and dynamic electricity pricing schemes, thereby contributing to broader grid stability and sustainability objectives [34,52].

1.3. The critical role of load prediction: From energy-saving operations to smart grid integration

Load forecasting has evolved from a purely operational tool for building management to a fundamental enabler of advanced energy services and grid-interactive building technologies. At its most basic level, accurate prediction of cooling and heating demands provides facility managers with the anticipatory intelligence necessary to optimize equipment scheduling, minimize energy consumption, and maintain occupant comfort within acceptable bounds [14,29,30]. However, the scope and significance of load prediction extend far beyond these traditional applications.

HVAC load forecasting offers several compelling advantages for building energy management: first, it enables proactive rather than reactive control, allowing systems to anticipate and prepare for changing conditions before they occur [11]; second, it facilitates optimal equipment scheduling and staging, reducing energy waste during transient periods and improving part-load performance; third, it supports participation in demand response programs and dynamic pricing schemes, generating both economic and grid-level benefits; and fourth, it provides the essential foundation for model predictive control strategies that can simultaneously optimize energy consumption, occupant comfort, and operating costs [6,19,81].

However, HVAC load forecasting also faces significant difficulties: first, the inherent complexity of building thermal dynamics, involving multivariable interactions, nonlinear relationships, and significant temporal variability across multiple timescales; second, the stochastic nature of key influencing factors, particularly occupant behavior and internal heat gains, which remain difficult to predict reliably; third, data quality challenges including sensor noise, missing values, and the "inaccurate load labels" problem identified by Tian et al. [63], where measured cooling capacity oscillates around true load values due to control system dynamics; fourth, the generalization challenge, as models trained on one building or climate zone often perform poorly when transferred to different contexts [9,65]; and fifth, the computational requirements of advanced methods, which may exceed the capabilities of standard building automation platforms for real-time implementation [78,94].

In the contemporary energy landscape, precise load forecasting serves as the cornerstone for implementing sophisticated demand-side management strategies, enabling buildings to function as responsive participants in smart grid ecosystems rather than passive consumers [34,41]. Advanced forecasting capabilities facilitate the integration of thermal energy storage systems, where predicted load profiles guide optimal charging and discharging cycles to exploit time-of-use electricity pricing differentials [1,4,52]. Furthermore, accurate load predictions enable the implementation of model predictive control

strategies that optimize system performance across extended time horizons, accounting for equipment constraints, comfort requirements, and economic objectives simultaneously [6,19,81].

The emergence of building-to-grid integration technologies has further elevated the importance of load prediction, as utilities increasingly rely on aggregated HVAC load forecasts for generation planning, grid balancing, and renewable energy integration [42,90]. This paradigm shift necessitates prediction methodologies that can deliver not only high accuracy but also quantified uncertainty estimates, enabling risk-informed decision-making across multiple stakeholders in the energy value chain [40,47].

Fig. 1 illustrates the evolution of HVAC load forecasting methods from 2010 to 2025, highlighting significant advancements in accuracy. Traditional statistical methods (e.g., MLR) achieved a MAPE of ~15%, while machine learning methods (e.g., SVM, RF) reduced it to ~8%. Deep learning techniques (e.g., LSTM, CNN) further improved accuracy to ~3%, and hybrid models (e.g., VMD + LSTM, Physics-informed AI) achieved MAPE values of approximately 2–5% under real-world operational conditions, with sub-1% values reported in certain simulation-based studies. Physics-informed AI approaches represent an emerging direction with preliminary results (e.g., 6.2% MAPE [84]) that have not yet matched the accuracy of mature hybrid models, though their value proposition lies primarily in improved generalization and interpretability rather than absolute accuracy. These advancements underscore the critical role of hybrid approaches in achieving superior forecasting performance.

1.4. Research motivation, scope, and structure

1.4.1. Research motivation

The scientific aim of this work is to provide a holistic examination of the transformative evolution of HVAC load prediction methodologies over the past decade, with particular emphasis on the convergence of data science, artificial intelligence algorithms, and modern control theory. The subject of the research was to systematically analyze 96 peer-reviewed publications to trace the progression from traditional statistical approaches to sophisticated hybrid frameworks, and to examine the complete pipeline from raw data preprocessing through intelligent system control.

Despite the extensive body of literature addressing HVAC load prediction, existing comprehensive reviews have predominantly focused on algorithmic comparisons while providing insufficient treatment of the critical preprocessing and feature engineering techniques that often determine model success [24,36]. Moreover, previous surveys have inadequately addressed the systematic integration of prediction methodologies with advanced control theories, missing the opportunity to examine the complete pipeline from raw data processing through intelligent system control.

Despite extensive research on HVAC load prediction, previous review studies exhibit several notable limitations:

(1) Insufficient treatment of data preprocessing: Existing reviews have predominantly focused on algorithmic comparisons while providing inadequate coverage of the critical preprocessing and feature engineering techniques that often determine whether a given model architecture can achieve its performance potential [24,36].

(2) Fragmented prediction-control pipeline: Previous surveys have inadequately addressed the systematic integration of prediction methodologies with advanced control theories (e.g., model predictive control, reinforcement learning), missing the opportunity to examine the complete pipeline from raw data processing through intelligent system control [94,95].

(3) Limited coverage of uncertainty quantification: Most prior reviews have not systematically evaluated how prediction models handle forecast uncertainty, despite its critical importance for robust control implementation and risk-informed decision-making [40,47].

(4) Absence of cross-disciplinary convergence analysis: Earlier works have generally treated data science, AI algorithms, and control theory as separate domains, lacking a holistic examination of their convergence and synergistic potential in building energy management.

(5) Insufficient coverage of specialized building types: Previous reviews have predominantly focused on commercial and residential buildings, with inadequate attention to industrial buildings, data centers, and other specialized facilities that present unique load forecasting challenges [33,74,82].

This study extends the existing research in HVAC load forecasting by: (I) Data science foundations: establishing data preprocessing and feature engineering as critical enabling foundations and providing the first systematic examination of how advanced signal decomposition, intelligent feature selection, and automated data quality assurance determine model performance potential across diverse algorithmic architectures; (II) Algorithmic convergence: tracing the complete evolutionary trajectory from traditional statistical methods through contemporary hybrid deep learning frameworks, revealing that the systematic integration of signal decomposition with hybrid architectures has been the primary driver of the most significant accuracy improvements; (III) Prediction-control integration: uniquely examining the convergence of prediction methodologies with advanced control theories including Model Predictive Control and deep reinforcement learning, demonstrating how accurate load forecasting enables energy savings of 10–30% with simultaneous comfort improvements; and (IV) Emerging paradigms and future outlook: critically assessing physics-informed AI, Large Language Model integration, and federated learning for building energy applications, providing maturity assessments and deployment timelines for each paradigm.

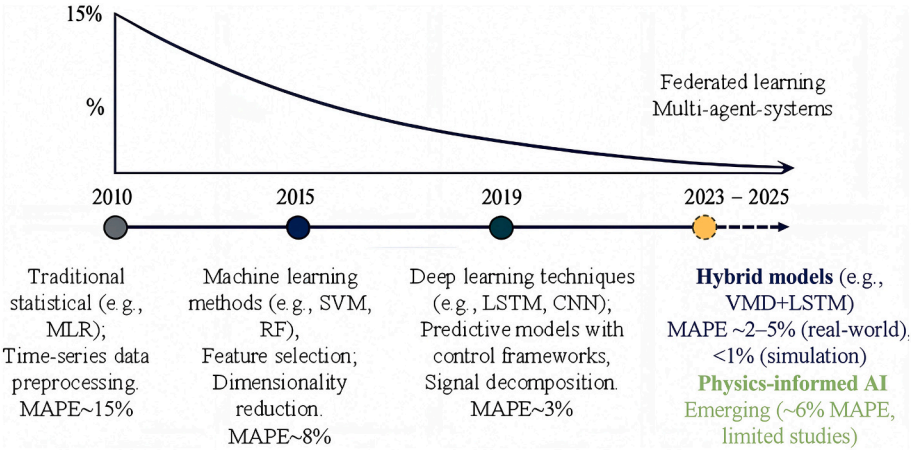


Fig. 1. Evolution of HVAC load forecasting methods and performance improvements (2010–2025).

1.4.2. Research scope

This comprehensive review systematically examines peer-reviewed research published primarily within the past decade, with concentrated focus on high-impact publications from the most recent five years that demonstrate significant methodological innovations. The scope encompasses data-driven prediction methodologies ranging from traditional machine learning approaches to cutting-edge deep learning architectures, with special attention to hybrid frameworks that combine multiple algorithmic paradigms. The analysis extends beyond pure prediction to examine the integration of forecasting results within advanced control strategies, including model predictive control, reinforcement learning, and emerging autonomous building management systems.

While the primary focus of this review is on HVAC thermal load forecasting (i.e., prediction of cooling and heating thermal demands), studies addressing whole-building energy consumption or HVAC electrical consumption are included where their methodological contributions, particularly in data preprocessing, feature engineering, or algorithmic innovation, are directly applicable to HVAC thermal load prediction. This distinction is maintained throughout the review to ensure terminological precision.

1.4.3. Structural organization

The subsequent sections of this review are strategically organized to provide both historical perspective and forward-looking insights. Section 2 establishes the fundamental importance of data science-driven preprocessing and feature engineering as the cornerstone of successful prediction systems. Section 3 traces the algorithmic evolution from traditional machine learning through contemporary deep learning approaches, emphasizing the emergence and proliferation of hybrid methodologies. Section 4 examines the critical transition from predictive modeling to intelligent control implementation, showcasing how advanced forecasting enables sophisticated building automation strategies. Finally, Section 5 identifies persistent challenges, emerging opportunities, and promising research directions, including a dedicated subsection on future outlook and recommendations (Section 5.3), that will shape the future trajectory of this rapidly evolving field.

2. The cornerstone of prediction: Data science-driven preprocessing and feature engineering

The foundation upon which any successful HVAC load prediction system rests is not merely the sophistication of its algorithmic core, but rather the systematic and intelligent treatment of raw operational data through advanced data science methodologies. Contemporary research has increasingly recognized that the quality of input data and the strategic engineering of predictive features serve as necessary prerequisites for model success, poor data quality can prevent even the most sophisticated algorithms from achieving acceptable performance [59,71]. However, it is important to distinguish between preprocessing as a necessary condition and preprocessing as a sufficient condition: the most significant accuracy advances have resulted from the synergistic combination of advanced preprocessing with architecturally superior hybrid models, rather than from either factor in isolation. This paradigm shift reflects a mature understanding that effective building energy forecasting requires a holistic approach to data management, where traditional “garbage in, garbage out” limitations are overcome through systematic application of data science principles.

2.1. Data acquisition and key influencing factor identification

The complexity of modern building energy systems necessitates a comprehensive approach to data acquisition that captures the multifaceted nature of thermal load generation.

The key factors influencing HVAC load forecasting can be systematically categorized into five groups:

(1) Meteorological factors: Weather data constitutes the primary external driving force, with outdoor dry-bulb temperature consistently emerging as the most influential predictor across diverse climatic conditions and building types [38,56,69]. Zhang et al. demonstrated that outdoor temperature correlations with cooling loads can reach values as high as 0.78, while the integration of additional meteorological parameters such as relative humidity, solar radiation, and wind speed provides incremental but significant improvements in prediction accuracy [60].

(2) Building-specific parameters: Building-specific parameters represent another critical data category, encompassing both static architectural characteristics and dynamic operational variables. Recent investigations by Zhao et al. revealed that incorporating building geometry parameters (including building shape coefficient, exterior wall area, and window-to-wall ratio) into prediction models can reduce prediction errors by up to 29.61% [53]. Meanwhile, indoor environmental data streams, particularly real-time measurements of occupancy patterns, equipment power consumption, and lighting schedules, have proven essential for capturing the stochastic nature of internal heat gains [26,33].

(3) Occupancy and behavioral factors: Occupancy patterns, internal heat gains from people and equipment, lighting schedules, and window/door opening behavior constitute significant yet stochastic influences on HVAC thermal loads. These factors are inherently difficult to predict but critically important for accurate forecasting, particularly in commercial and institutional buildings with variable usage patterns [12,33,48].

(4) Temporal and calendar factors: Time of day, day of week, holiday schedules, seasonal variations, and lag effects arising from building thermal inertia all exert systematic influences on load profiles. The exploitation of these temporal patterns through appropriate feature engineering has been shown to significantly improve prediction accuracy [14,55].

(5) Operational and system factors: HVAC operational states, chilled / hot water supply temperatures and flow rates, equipment start-up and shutdown transients, system maintenance status, and equipment degradation over time all influence the relationship between thermal demand and system response. An important consideration in this category concerns the Coefficient of Performance (COP) of cooling and heating equipment, which varies dynamically with operating conditions including chilled water supply temperature, condenser water temperature, outdoor ambient conditions, and part-load ratio [2,54]. Since most HVAC load prediction models target thermal loads (cooling or heating demands), the translation of predicted thermal loads to electrical energy consumption, which is essential for grid integration, demand response, and cost optimization applications, requires accurate COP modeling. Several studies have addressed this challenge through different approaches: (i) direct electrical load prediction that implicitly accounts for COP variations through data-driven modeling [43,66]; (ii) cascaded frameworks where thermal load predictions are coupled with equipment performance models that capture COP dependencies on operating conditions [54,87]; and (iii) physics-informed approaches that embed thermodynamic relationships governing COP behavior into the prediction framework [84,93]. The dynamic nature of COP, which can vary by 30–50% across typical operating ranges, introduces a significant source of uncertainty when converting thermal load forecasts to electrical consumption predictions, highlighting the need for integrated prediction-performance modeling approaches that capture these equipment-level dynamics.

Industrial buildings present unique load forecasting challenges distinct from commercial or residential settings. These facilities are characterized by process-driven thermal loads, specialized equipment heat dissipation, and production-schedule-dependent occupancy patterns that create complex, non-stationary load profiles. Zhang et al. [33] developed and evaluated cooling load prediction models specifically for a factory workshop environment, demonstrating that industrial load characteristics, including large internal heat gains from manufacturing

equipment and irregular operational schedules, necessitate tailored feature selection and model architectures. Zuo et al. [74] addressed the prediction of heating and cooling loads for factory mushroom houses using Empirical Wavelet Transform (EWT) decomposition, illustrating how agricultural-industrial hybrid facilities require specialized signal processing approaches. Furthermore, Zhao and Fan [82] proposed short-term load forecasting methods specifically designed for industrial buildings based on signal decomposition and composite prediction models, highlighting the importance of capturing the multi-scale temporal dynamics unique to manufacturing environments. These studies collectively underscore that industrial building load prediction remains an underexplored yet important domain relative to the commercial and residential building sectors.

The emergence of Internet of Things technologies in building automation has greatly expanded the availability of high-frequency operational data from HVAC equipment itself. He et al. pioneered the utilization of real-time system parameters, such as including water temperatures, flow rates, and equipment operational states, as observational variables in closed-loop prediction frameworks, achieving accuracy improvements of 42.50% over conventional open-loop approaches [51]. This evolution toward comprehensive sensor integration reflects the broader transformation of buildings into data-rich environments where traditional limitations in data availability are being systematically eliminated.

2.2. Data cleaning and normalization methodologies

The transition from raw sensor measurements to analysis-ready datasets represents a critical yet often underestimated component of the prediction pipeline. Real-world building operational data invariably contains anomalies, outliers, and systematic errors that can severely compromise model performance if left unaddressed. Shao et al. established that implementing rigorous data quality control procedures (including normality testing via Q-Q plots and outlier removal using $\mu \pm 2\sigma$ confidence intervals) can reduce prediction errors from 4.83% to 2.22% in hotel building energy forecasting applications [42].

Missing data presents another pervasive challenge in building energy datasets, frequently arising from sensor malfunctions, communication failures, or maintenance activities. Song et al. addressed this challenge through an innovative approach combining bidirectional Long Short-Term Memory networks for intelligent missing value imputation, demonstrating that their method could accurately reconstruct up to 15.68% missing data while maintaining prediction model integrity [89]. Alternative strategies have included Kalman filtering for anomaly correction [73] and locally weighted scatterplot smoothing for noise reduction [57], each offering distinct advantages depending on the underlying data characteristics.

Normalization and standardization procedures have proven equally critical for ensuring optimal model performance, particularly when dealing with heterogeneous data sources spanning multiple orders of magnitude. The systematic application of *Z*-score normalization or min-max scaling not only accelerates model convergence but also prevents certain features from dominating the learning process due to their numerical scale rather than their predictive importance [2,15].

2.3. Advanced feature selection and dimensionality reduction techniques

The exponential growth in available building data streams has created new challenges related to the “curse of dimensionality”, where excessive feature inclusion can paradoxically degrade model performance through increased noise, computational overhead, and overfitting risks. Contemporary research has responded with increasingly sophisticated feature selection methodologies that balance comprehensive information capture with model parsimony.

Filter-based approaches have demonstrated strong effectiveness in preliminary feature screening, with Pearson and Spearman correlation coefficients serving as foundational tools for identifying linear and monotonic relationships between predictors and target variables [1,4,27,53]. However, these traditional metrics prove insufficient for capturing complex nonlinear associations that characterize building energy systems. Yang et al. advanced this field by introducing Maximum Information Coefficient analysis, which successfully identified nonlinear feature relationships that conventional correlation analysis failed to detect, resulting in 40.26–74.18% reductions in prediction errors [77].

Wrapper and embedded methods have gained prominence for their ability to evaluate feature subsets within the context of specific prediction algorithms. The Random Forest-based Recursive Feature Elimination approach proposed by Gao et al. exemplifies this trend, systematically reducing 14 potential input variables to an optimal subset of 8 features while simultaneously improving model accuracy [50]. Similarly, the Boruta algorithm implementation by Zuo et al. achieved a 14.72% reduction in prediction errors through intelligent feature selection that outperformed traditional correlation-based approaches [74].

Principal Component Analysis and its variants continue to serve crucial roles in addressing multicollinearity issues while preserving essential information content. Guo et al. demonstrated that PCA could capture over 95.5% of variance in meteorological data while reducing input dimensionality from four to one principal component, significantly improving model stability [8]. More recently, Lv et al. extended this concept through the integration of Partial Auto-correlation Function analysis with PCA, achieving enhanced feature engineering specifically tailored for time-lagged thermal relationships in building systems [55].

2.4. Signal decomposition strategies for non-stationary time series

The evolution of signal decomposition methodologies in HVAC load prediction reflects a sophisticated understanding of the multi-scale temporal dynamics inherent in energy consumption patterns. Early applications primarily relied on Empirical Mode Decomposition (EMD), which, despite its adaptive nature, suffered from mode mixing and endpoint effects that limited its practical effectiveness [30,34]. The field subsequently progressed through Ensemble Empirical Mode Decomposition (EEMD) to address these limitations, before culminating in the development of Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) [53,73].

Variational Mode Decomposition has emerged as the current gold standard for building energy time series analysis, offering superior performance through its solid mathematical foundation and robust noise resilience [45,86]. Gao et al. demonstrated that VMD-based decomposition, when combined with Markov chain error correction, could achieve determination coefficients of 0.96 while reducing prediction errors by over 19% compared to conventional approaches [45]. The success of VMD stems from its ability to simultaneously decompose signals into intrinsic mode functions while minimizing bandwidth constraints, thereby producing more stable and predictable sub-components.

Wavelet Transform methodologies have carved out a specialized niche in applications requiring precise frequency domain analysis of load patterns. The comprehensive framework developed by Zhao and Liu utilized Dmeyer wavelet functions for two-level decomposition, successfully separating load signals into periodic low-frequency components and stochastic high-frequency disturbances [22]. This approach enabled targeted modeling strategies where linear regression techniques addressed trend components while support vector machines handled random fluctuations, achieving superior performance compared to single-model approaches.

Recent innovations have pushed beyond traditional decomposition boundaries, with researchers exploring hybrid decomposition strategies that combine multiple techniques for enhanced signal processing. The Improved CEEMDAN approach proposed by Zhou et al. integrated Particle Swarm Optimization for automatic parameter selection, followed by secondary VMD processing of complex high-frequency components

[86]. This multi-stage decomposition achieved remarkable accuracy with determination coefficients reaching 0.9901, establishing new benchmarks for signal preprocessing in building energy applications.

2.5. Emerging data processing paradigms

The frontier of HVAC load prediction has witnessed the emergence of novel data processing paradigms that transcend traditional statistical boundaries, incorporating concepts from information theory, fuzzy logic, and artificial intelligence to create more intelligent and adaptive preprocessing frameworks. Clustering-based operational mode identification represents one such paradigm, where unsupervised learning techniques systematically categorize historical operational data into distinct behavioral patterns [41,70]. Zhang et al. pioneered this approach by employing K-means clustering to identify five distinct load modes based on daily profile characteristics, subsequently achieving a 10% improvement in prediction accuracy through mode-specific modeling strategies [33].

Information granulation techniques have introduced sophisticated approaches to uncertainty quantification and interval prediction, moving beyond deterministic point forecasts to provide probabilistic assessments of future load ranges. Wang et al. applied fuzzy information granulation using triangular fuzzy particles to transform complex daily load profiles into simplified three-parameter representations, enabling Support Vector Machine models to forecast both load magnitude and fluctuation ranges with exceptional accuracy [5]. This methodology demonstrated high precision precision with actual load measurements falling entirely within predicted intervals, validating the practical utility of granulation-based approaches.

The integration of expert knowledge through analytical hierarchy processes has emerged as another promising direction for enhancing data-driven models with domain expertise. Tan et al. developed an innovative framework combining Analytic Hierarchy Process with neural network prediction, where expert-derived weights for influence factors were systematically quantified and embedded into the prediction model structure [3]. This approach achieved Mean Absolute Relative Error of only 5.40%, demonstrating that the strategic incorporation of human expertise can significantly enhance purely data-driven methodologies.

Table 1 presents a comparative summary of the advanced data preprocessing techniques reviewed in this section, encompassing signal decomposition, feature selection, clustering-based processing, information granulation, and expert knowledge integration, with respect to their representative methods, key advantages, primary limitations, typical applications, and performance impacts.

The flowchart in Fig. 2 illustrates the data-driven workflow for HVAC load forecasting, encompassing six key steps: data acquisition, cleaning, normalization, feature selection, dimensionality reduction, and signal decomposition. By leveraging advanced techniques like isolation forest, SHAP, PCA, and VMD, the process can enhance data quality, perform efficient dimensionality reduction, and extract meaningful features. The resulting model-ready dataset ensures high feature relevance and data integrity, providing a robust foundation for accurate load forecasting and subsequent prediction model input.

3. The evolution of core algorithms: From traditional machine learning to deep learning

The algorithmic landscape of building HVAC load prediction has undergone a significant transformation over the past two decades, characterized by an evolutionary progression from simple statistical models through sophisticated machine learning frameworks to cutting-edge deep learning architectures. This evolution reflects not merely technological advancement, but rather a fundamental shift in how researchers conceptualize and approach the inherently complex, nonlinear, and temporally dependent nature of building energy consumption patterns.

An important methodological caveat must be noted when interpreting the accuracy comparisons presented in this section. The HVAC load forecasting literature exhibits considerable heterogeneity in experimental design: studies differ in building types (office, residential, commercial, industrial), climate zones (tropical, temperate, continental, arid), prediction horizons (15-min to 24-h ahead) [18], data collection periods (weeks to years), and validation strategies (random split, temporal split, cross-validation, independent test sets). Consequently, the accuracy figures reported across different studies serve as indicative

Table 1
Comparative analysis of advanced data preprocessing techniques in HVAC load prediction.

Technique category	Representat-ive methods	Key advantages	Primary limitations	Typical applications	Performance impact
Signal decomposi-tion	VMD [45,86], CEEMDAN [53,73], Wavelet transform [22,34]	Effective handling of non-stationarity; Improved signal regularity; Enhanced predictability	Computation-al overhead; Parameter sensitivity; Potential over-decomposi-tion	Non-stationary load series; Peak load analysis; Multi-scale forecasting	15–50% error reduction
Feature selection	Correlation analysis [1,4], Random forest-RFE [50], Boruta algorithm [74]	Dimensiona-lity reduction; Overfitting prevention; Computation-al efficiency	Information loss risk; Algorithm dependency; Limited nonlinear capture	High-dimensional datasets; Multicolline-arity issues; Real-time applications	10–30% accuracy improvement
Clustering-based processing	K-means mode identification [33,41], Fuzzy C-means [70]	Operational pattern recognition; Targeted modeling; Improved generaliza-tion	Cluster number selection; Interpretabili-ty challenges; Mode transition handling	Multi-mode buildings; Seasonal variations; Occupancy-driven loads	8–15% precision enhancement
Information granulation	Fuzzy granulation [5], Rough set theory [43]	Uncertainty quantifica-tion; Interval prediction; Noise resilience	Computatio-nal complexity; Parameter tuning; Limited adoption	Probabilistic forecasting; Risk assessment; Decision support	20–40% uncertainty reduction
Expert knowledge integration	Analytic hierarchy process [3], Entropy weight method [68]	Domain expertise utilization; Interpretabili-ty enhancement; Robustness improvement	Subjectivity concerns; Scalability limitations; Expert availability	Complex systems; Multi-criteria evaluation; Hybrid modeling	12–25% stability improvement

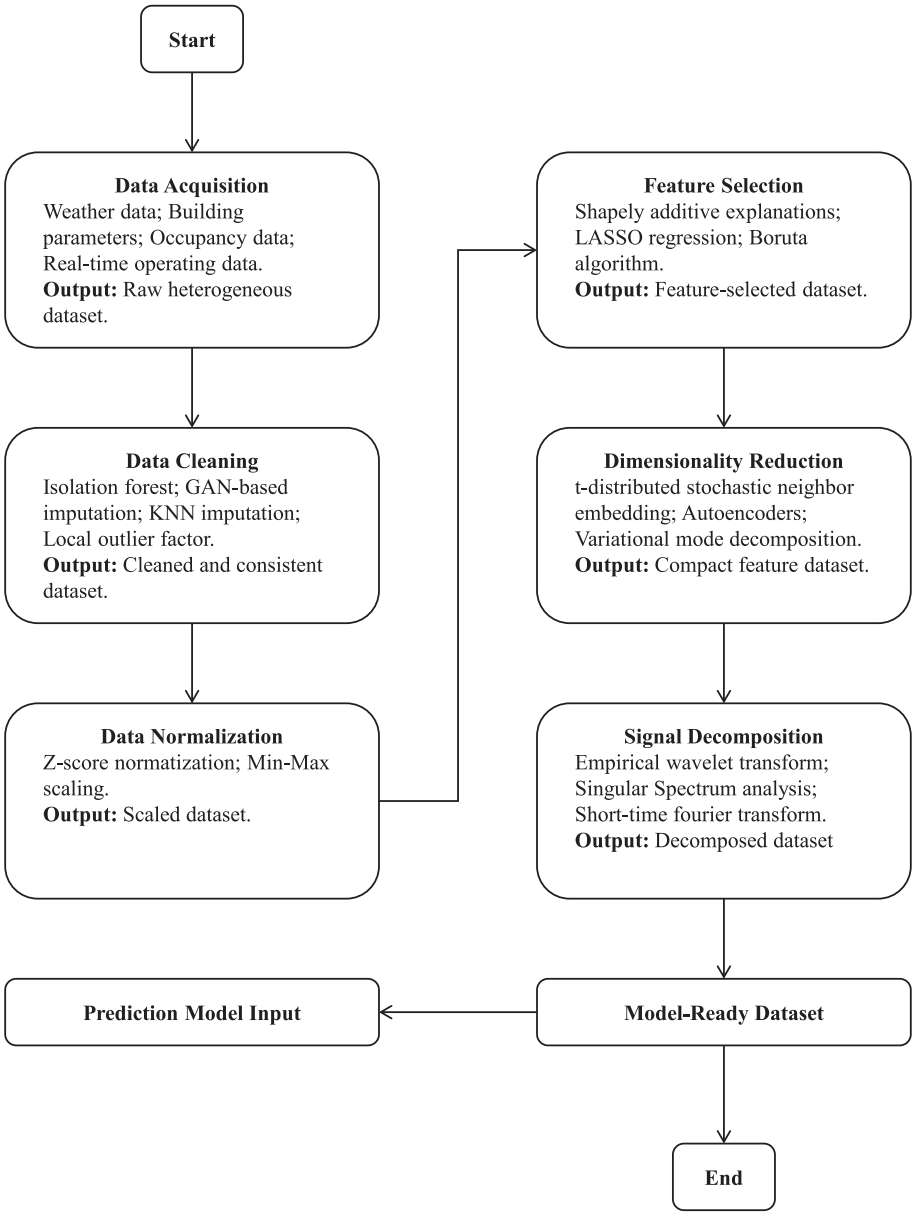


Fig. 2. Comprehensive data processing workflow for HVAC load forecasting.

performance ranges rather than strictly comparable benchmarks. Meaningful performance comparison requires evaluation on common datasets under standardized protocols, a community-wide effort that remains an important gap in the field.

3.1. The traditional machine learning era: Establishing foundations

During the foundational period of data-driven HVAC load prediction (approximately 2010–2018), researchers primarily relied upon well-established machine learning paradigms that offered reasonable accuracy while maintaining computational tractability and interpretability. Multiple Linear Regression and related statistical approaches have served as foundational tools in building load estimation for several decades, with well-established applications dating back to at least the 1970s. In the context of the data-driven prediction literature reviewed here, Guo et al. demonstrated that carefully engineered MLR models, when enhanced with Principal Component Analysis and dynamic correction mechanisms, could still achieve competitive performance with Mean Absolute Relative Error below 8% [8]. However, the fundamental limitation of linear models in capturing the complex nonlinear relationships inherent in building thermal dynamics quickly became apparent, driving researchers toward more sophisticated alternatives.

Support Vector Machines gained substantial traction during this era due to their exceptional performance in small-sample scenarios and robust generalization capabilities. The comprehensive analysis by Ahmad and Chen revealed that optimized SVR models could achieve Mean Absolute Percentage Error as low as 0.405% for 7-day cooling load forecasting, significantly outperforming traditional linear approaches [25]. The success of SVM-based methods largely stemmed from their ability to handle high-dimensional feature spaces through kernel transformations while maintaining computational efficiency, making them particularly attractive for real-time building management applications. Complementary soft computing approaches incorporating fuzzy inference and hybrid intelligent methods have similarly demonstrated strong predictive capabilities for residential building energy performance prediction [17].

Ensemble learning methodologies represented the sophisticated

frontier of traditional machine learning, with Random Forest, Gradient Boosting, and their variants demonstrating robust performance across diverse building types and climatic conditions. Zhang et al. established that Gradient Boosting Decision Trees could achieve determination coefficients exceeding 0.98 under both general and high-load operational modes, with Mean Absolute Error reductions of approximately 30% compared to individual decision trees [44]. The inherent ability of ensemble methods to combine multiple weak learners into powerful predictors while providing natural uncertainty quantification made them invaluable tools for practical building energy management applications. Data-driven prediction approaches have also been successfully extended to specialized HVAC configurations including Variable Refrigerant Flow (VRF) heating systems, where energy cost prediction models have been developed to support system-level operational optimization [21].

3.2. The deep learning revolution: Unprecedented capability and complexity

The introduction of deep learning methodologies marked a significant shift in HVAC load prediction, offering enhanced capability to model complex, high-dimensional relationships while automatically extracting hierarchical features from raw data. Artificial Neural Networks experienced a renaissance through deeper architectures and advanced training algorithms, with Fan et al. demonstrating that deep learning approaches could achieve determination coefficients of 0.963 for commercial building cooling load prediction [15], while Deb et al. demonstrated ANN-based diurnal cooling load forecasting for institutional buildings [10], collectively confirming deep learning's superiority over conventional machine learning alternatives across diverse building types.

Recurrent Neural Networks revolutionized the treatment of temporal dependencies in building energy data, with Long Short-Term Memory networks emerging as the dominant architecture for sequence modeling applications. Zhou et al. conducted research demonstrating that LSTM networks could predict air-conditioning electrical energy consumption with remarkable accuracy using only historical consumption data as input, achieving Mean Absolute Percentage Error reductions of 16.31% compared to traditional ARIMA models and 36.61% compared to backpropagation networks [43]. LSTM's success stemmed from its sophisticated gating mechanisms that enabled selective information retention and forgetting, effectively addressing the vanishing gradient problem that plagued earlier recurrent architectures. Alternative recurrent formulations, including Nonlinear AutoRegressive with eXogenous inputs (NARX) neural networks optimized through predictive optimization strategies, have also demonstrated competitive performance for building cooling load forecasting [35].

Convolutional Neural Networks found innovative applications in building energy prediction through their ability to extract local patterns and spatial relationships from multi-dimensional data. The pioneering work of Li et al. introduced one-dimensional CNN architectures for building cooling load forecasting, where convolutional layers effectively captured local temporal features before feeding processed information to subsequent LSTM layers [62]. This hybrid CNN-LSTM approach achieved exceptional performance with Root Mean Square Error of 0.01 and determination coefficient of 0.98, establishing new benchmarks for deep learning applications in building energy systems.

Transformer architectures represent an advanced frontier in contemporary deep learning applications, offering strong capability in modeling long-range dependencies through self-attention mechanisms. Li et al. successfully adapted Transformer models for HVAC load prediction, demonstrating that attention-based architectures could achieve determination coefficients of 0.98 with uncertainty ranges of only ±11%, significantly outperforming LSTM-based alternatives [62]. Transformer models can process entire sequences in parallel while dynamically weighting different temporal positions. This allows them to capture both short-term fluctuations and long-term seasonal patterns effectively.

Graph Neural Networks have emerged as a promising approach for modeling multi-zone buildings and district energy systems, where spatial relationships between different building zones or equipment components play crucial roles in overall system behavior. Chen et al. pioneered the application of knowledge-embedded Graph Convolutional Networks, achieving Mean Absolute Percentage Error of 14.92% through dynamic modeling of feature relationships and expert knowledge integration [61]. This approach represents a significant advancement in capturing the complex interdependencies that characterize modern building energy systems.

3.3. The hybrid model revolution: Synergistic integration for enhanced performance

Contemporary HVAC load prediction has been dominated by the emergence of hybrid modeling strategies that synergistically combine multiple algorithmic paradigms to achieve performance levels unattainable by individual approaches. These hybrid frameworks typically manifest in two primary configurations: optimization-enhanced prediction models and decomposition-integrated prediction systems.

Optimization-enhanced hybrid models leverage metaheuristic algorithms to systematically optimize the hyperparameters, weights, or structural configurations of base prediction models. Guo et al. exemplified this approach through their Improved Genetic Algorithm-optimized Deep Belief Network, achieving prediction accuracy of 96.52% with 14.32% reduction in model energy consumption compared to conventional approaches [1]. The success of such frameworks lies in their ability to overcome the random initialization problems that plague neural networks while ensuring global optimization rather than local minima convergence.

Decomposition-integrated prediction systems represent perhaps the most sophisticated contemporary approach, combining advanced signal processing with multiple specialized prediction models. The comprehensive framework developed by Zhou et al. integrated Improved Variational Mode Decomposition with Whale Optimization Algorithm-enhanced Least Squares Support Vector Machines, achieving determination coefficients of 0.9901 and Coefficient of Variation Root Mean Square Error of only 0.0344 [86]. This approach demonstrates the power of “divide-and-conquer” strategies where complex load signals are decomposed into simpler components, each modeled with algorithms specifically suited to their characteristics.

Ensemble learning at scale has evolved beyond traditional bagging and boosting to incorporate sophisticated meta-learning strategies. Pachauri and Ahn developed a weighted aggregated ensemble model combining Gaussian Process Regression with Least Squares Boosting, optimized through Marine Predator Optimization algorithm [57]. Their model achieved Root Mean Square Error reductions of 12.4–70.7% compared to individual algorithms, while Multi-Criteria Performance Index rankings confirmed state-of-the-art performance across multiple evaluation metrics.

Despite the impressive accuracy figures reported by hybrid models, several critical concerns temper their practical significance. The majority of these models have been validated on single buildings or limited datasets, with very few studies demonstrating consistent performance across diverse building types and climatic conditions. The reproducibility of reported results is also questionable, as most studies do not provide open-source implementations or publicly accessible datasets, making independent verification difficult.

While hybrid decomposition-prediction models have demonstrated superior accuracy in research settings, several practical deployment challenges warrant careful consideration. First, computational complexity scales multiplicatively: a VMD-LSTM hybrid model requires sequential execution of the decomposition algorithm, individual model training for each decomposed component, and result reconstruction, a

pipeline that can be orders of magnitude more computationally expensive than a single-model approach. Zhou et al. [86] reported that their IVMD-WOA-LSSVM framework, while achieving exceptional accuracy, required substantially more computation time than individual baseline models. Second, hyperparameter sensitivity is amplified in hybrid frameworks because each component introduces its own set of tunable parameters: the number of decomposition modes in VMD, the population size and iteration count in metaheuristic optimizers, and the architecture-specific parameters of each sub-model must all be jointly optimized. Yang et al. [75] acknowledged that the improved sparrow search algorithm, while effective, introduced additional tuning complexity. Third, the suitability of these complex pipelines for real-time applications, where predictions must be generated within seconds, has not been rigorously demonstrated by most studies. He et al. [78] specifically addressed the small-sample, real-time scenario using permutation entropy and temporal convolutional networks as a computationally lighter alternative, suggesting that the most complex hybrid models may not be appropriate for all deployment contexts. Building practitioners must therefore carefully evaluate whether the marginal accuracy improvements offered by complex hybrid models justify the substantially increased computational overhead, maintenance complexity, and expertise requirements relative to simpler, more robust alternatives.

A critical limitation of the current literature is the absence of standardized benchmark datasets and evaluation protocols for HVAC load prediction. Unlike domains such as computer vision (ImageNet) or natural language processing (GLUE), the HVAC load forecasting community lacks widely adopted reference datasets that enable fair cross-method comparisons. This deficiency means that the accuracy ranges reported in Table 2 should be interpreted with significant caution, and the establishment of community-wide benchmark standards represents an urgent research priority.

The trajectory from traditional machine learning through deep learning to sophisticated hybrid approaches reflects not merely technological progress, but a fundamental evolution in understanding the multi-faceted nature of building energy prediction challenges. Each algorithmic generation has contributed essential insights while simultaneously revealing new layers of complexity that drive continued innovation in this rapidly advancing field.

Fig. 3 presents a comparative performance analysis of prediction algorithms for HVAC load forecasting. All axes are normalized to a 1–5 scale where higher values indicate more desirable performance. “Computational Efficiency” and “Data Efficiency” represent inverted scales of computational complexity and training data demand, respectively.

4. From prediction to control: Application and closing the loop with advanced theories

The transformation of building energy management from reactive operational paradigms to proactive, intelligence-driven control strategies represents one of the most significant advances in contemporary

Table 2
Evolution and performance comparison of core prediction algorithms.

Algorithm category	Representati-ve models	Period of prominent application in reviewed data-driven literature	Reported MAPE range in reviewed literature	Primary advantages	Notable limitations	Key contextual factors
Linear models	MLR [8], ARX [16,30], ARIMA [43]	Pre-2015 (foundation-al)[a]	Approximately 8–15%	Interpretabili-ty; Fast computation; Low data requirements	Limited nonlinearity handling; Poor complex pattern capture	Small-to-moderate datasets; Single buildings; Well-characterized systems
Traditional ML	SVM/SVR [5,32], Random forest [44,50], XGBoost [39,49]	2015–2019	3–8%	Robust generaliza-tion; Ensemble benefits; Good small-sample performance	Manual feature engineering; Limited temporal modeling	Moderate datasets; Various building types; Feature engineering required
Basic deep learning	BPNN [2,3], DBN [1,28], Basic RNN [43]	2016–2020	2–6%	Automatic feature extraction; Nonlinear capability; Hierarchical learning	Training complexity; Overfitting risk; Limited sequence modeling	Medium-to-large datasets; Mostly single-building validation
Advanced deep learning	LSTM [43,67], CNN [62,79], GRU [85,91]	2019–2023	1–4%	Superior temporal modeling; Long-term dependen-cies; Multi-scale feature extraction	High computation-al cost; Large data requirements; Black-box nature	Large datasets; Some multi-building studies; Temporal modeling focus
Frontier architec-tures	Transformer [62,83], GNN [61,87], TCN [72,89]	2022-Present	0.5–3%[b]	Attention mechanisms; Complex relationship modeling; Parallel processing	Implementa-tion complexity; Limited interpretabili-ty; Domain adaptation needs	Limited real-world validation; Mostly simulation or single-building studies
Hybrid models	VMD-LSTM [45], GA-DBN [1], Ensemble methods [57,64]	2020-Present	0.2–2%[c]	Synergistic benefits; Robust performance; Multi-objective optimization	Increased complexity; Parameter sensitivity; Integration challenges	Varied datasets; Primarily simulation and single-building validation; Complex pipelines

Note: The reported MAPE ranges are NOT directly comparable across algorithm categories, as they are drawn from studies with different datasets, building types, climate zones, prediction horizons, data collection periods, and validation protocols. These values serve as indicative ranges observed in the reviewed literature rather than standardized performance benchmarks. Studies using simulation-generated data typically report lower MAPE values than those using real operational measurements.

[a] Linear regression methods have been applied in HVAC load estimation for several decades. The date ranges indicate periods of prominent appearance in the reviewed data-driven literature, not original development periods.

[b] Frontier architectures have been validated on fewer and often more challenging datasets compared to established methods, which may partially account for their wider reported MAPE ranges.

[c] Sub-1% MAPE values have been reported primarily in simulation-based studies. Field-deployed accuracy is constrained by sensor measurement uncertainties (typically 1–5%) [2,63].

Table 3
Integration of load forecasting with advanced control strategies in building energy management.

Control approach	Prediction integration	Key performance metrics	Representat-ive studies	Primary benefits	Implementa-tion challenges
Model predictive control	Day-ahead load forecasting; Real-time prediction updates	20–30% energy savings; 80% redundancy reduction	Ma et al. [6]; Wang et al. [31]; Zhao et al. [81]	Optimal scheduling; Constraint handling; Multi-objective optimization	Model complexity; Computa-tional requirements; Uncertainty quantification
Deep reinforce-ment learning	State space enhancement; Predictive state variables	10–15% cost reduction; 79% comfort improvement	Du et al. [46]; Wang et al. [90]	Autonomous adaptation; No model requirements; Continuous learning	Training time; Sample efficiency; Safety concerns
Multi-agent coordination	Distributed load forecasting; Community-level prediction	12.5% consumption reduction; 10% peak demand reduction	Ghofrani et al. [34]	Scalability; Collective optimization; Grid integration	Communica-tion overhead; Coordination complexity; Privacy issues

Table footnote: The control approaches listed in this table represent strategies that directly govern HVAC system operation through control actions. Emerging model training and development paradigms (transfer learning, federated learning, automated machine learning), which address deployment challenges but do not directly generate control actions, are categorized separately in Table 4.

Table 4
Emerging learning paradigms for addressing deployment challenges in building load prediction.

Control approach	Prediction integration	Key performance metrics	Representat-ive studies	Primary benefits	Implementa-tion challenges
Transfer learning	Cross-domain prediction models; Few-shot adaptation	63% error reduction with 1% target data	Kim et al. [65]; Zhang et al. [80]	Data efficiency; Rapid deployment in data-scarce buildings; Domain adaptation	Source-target mismatch; Negative transfer risk; Knowledge transfer validation
Federated learning	Privacy-preserving collaborative training; Multi-building cooperation	Comparable to centralized training	Song et al. [88]	Privacy preservation; Collaborative learning; Distributed intelligence	Communication costs; Heterogeneous data handling; Convergence guarantees
Auto-ML integration	Automated pipeline optimization; Self-tuning prediction systems	1–19% accuracy improvement over manual approaches	Zhang et al. [59]; Lu et al. [71]	Accessibility for non-experts; Automation of model selection; Reduced expertise requirements	Limited customiza-tion; Black-box operation; Domain-specific adaptation needs

Table footnote: The learning paradigms listed in this table are model training and development frameworks that address practical barriers to deploying advanced prediction models, such as data scarcity, privacy constraints, and expertise requirements. They are distinct from the control strategies in Table 3 in that they primarily concern how prediction models are constructed, trained, and maintained, rather than how HVAC control actions are generated and executed. Current maturity levels for HVAC applications vary: transfer learning has been demonstrated in multiple building energy studies [65,80]; federated learning has shown initial feasibility [88] but faces substantial challenges related to non-IID data distributions and communication overhead; AutoML frameworks have been evaluated for building energy prediction [59,71] but domain-specific adaptation remains limited.

building automation. This evolution has been fundamentally enabled by the convergence of accurate load forecasting capabilities with sophisticated control theories, creating closed-loop systems that anticipate future conditions rather than merely responding to current deviations. The integration of predictive intelligence with advanced control frameworks has unlocked substantial opportunities for energy optimization, demand response participation, and autonomous building operation.

4.1. Model predictive control integration: Anticipatory optimization through forecasting

Combining load forecasting and Model Predictive Control has emerged as a significant advancement in building energy management, fundamentally altering how HVAC systems respond to dynamic thermal demands. Unlike traditional feedback control systems that react to temperature deviations after they occur, MPC frameworks leverage predicted load profiles to optimize system operation across extended time horizons while explicitly accounting for system constraints, economic objectives, and comfort requirements [6,19,81]. A comprehensive evaluation of nearly 100 high-impact MPC studies over the past decade reveals that MPC research is progressively shifting from simple white-box setups toward gray- and black-box models paired with metaheuristic or hybrid solvers, increasingly leveraging machine learning for forecasting and multi-objective optimization, although significant gaps remain in robustness validation, standardized benchmarking, and real-world deployment [94].

Ma et al. established the theoretical framework for Stochastic Model Predictive Control in building HVAC systems, where load prediction uncertainty was explicitly incorporated into the optimization formulation through probabilistic constraints [6]. Their research demonstrated that accounting for prediction uncertainty through Monte Carlo simulation and scenario-based optimization could achieve substantial energy savings while maintaining robust comfort guarantees, even when load predictions deviated from actual values. This pioneering approach recognized that prediction accuracy, while important, was less critical than the systematic quantification and accommodation of prediction uncertainty within the control framework.

Subsequent developments have focused on practical implementation strategies that bridge the gap between theoretical MPC formulations and real-world building systems. Wang et al. developed a comprehensive predictive optimization framework for chiller plants that integrated ensemble-based load forecasting with genetic algorithm optimization, achieving 80.5% reduction in redundant energy consumption compared to conventional hourly optimization strategies [31]. Their research demonstrated that the economic benefits of predictive control directly

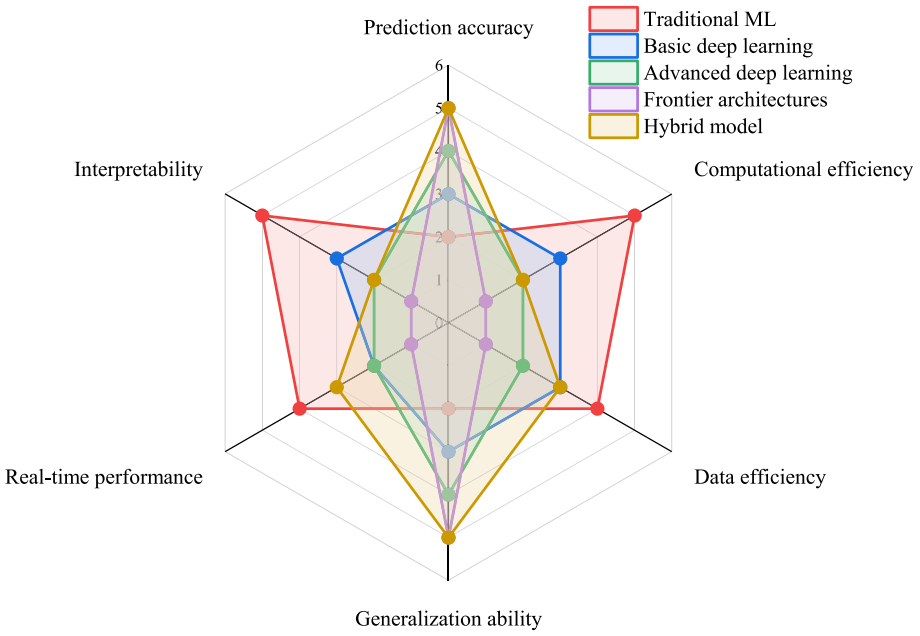


Fig. 3. Comparative performance analysis of prediction algorithms for HVAC load forecasting.

correlate with forecasting accuracy, establishing a quantitative relationship between prediction precision and operational savings. Furthermore, the translation from predicted thermal loads to optimal control actions must account for the dynamic COP of cooling and heating equipment, as discussed in Section 2.1, since identical thermal load predictions may result in substantially different electrical energy consumption depending on equipment operating conditions.

Ice storage and thermal energy storage integration has further amplified the importance of accurate load prediction in MPC applications. Zhao et al. developed a sophisticated multi-objective MPC framework for ice-storage air-conditioning systems, where day-ahead load forecasts guided optimal ice production scheduling while real-time predictions enabled dynamic operational adjustments [81]. Their results revealed that MPC strategies could achieve 25% energy consumption reduction and 20.9% operational cost savings compared to rule-based control, with actual performance deviating from theoretical optimum by merely 0.2% in energy consumption and 0.4% in operational costs.

A critical insight emerging from recent research is that the relationship between prediction accuracy and control performance is not monotonically linear. Improvements in forecasting precision do not always translate proportionally, or even positively, into better operational outcomes. Wang et al. [76] provided compelling evidence of this phenomenon in the context of an ice-based thermal energy storage system, revealing a fundamental disconnect between conventional prediction accuracy metrics (MAPE, RMSE) and actual control performance metrics (energy consumption, cost savings). Their research demonstrated that a model with superior statistical accuracy did not necessarily yield better control outcomes, because the temporal distribution and directionality of prediction errors matter more for control performance than aggregate error statistics. For instance, systematic overprediction of cooling load leads to conservative control strategies with excessive pre-cooling, while systematic underprediction results in comfort violations. In both scenarios, low MAPE may coexist with poor operational outcomes.

This finding has profound implications for the research community's evaluation framework. In MPC applications, prediction errors interact with the optimization horizon, constraint handling, and cost function design, creating complex error propagation pathways that simple accuracy metrics cannot capture. Similarly, in reinforcement learning, the state representation that incorporates load predictions may be more sensitive to prediction bias than to prediction variance. Ma et al. [6] demonstrated in their stochastic MPC framework that systematically quantifying and accommodating prediction uncertainty through probabilistic constraints was more important for robust control performance than minimizing point prediction error. These findings suggest that future research should evaluate prediction models not only on traditional statistical metrics but also on their downstream control performance through integrated simulation or field testing.

4.2. Reinforcement learning in dynamic control: Learning optimal strategies through environmental interaction

The application of Reinforcement Learning in building HVAC control represents a paradigm shift toward autonomous, adaptive systems that learn optimal control policies through direct interaction with the building environment, without requiring explicit mathematical models of system dynamics. This approach has proven particularly valuable in complex, multi-zone residential and commercial buildings where traditional control strategies struggle to accommodate the stochastic nature of occupant behavior and varying operational conditions. A systematic review of 73 real-world ML-based building energy management deployments reveals that RL is gaining traction through safer offline-online learning strategies, with real-world deployments following a mature algorithmic blueprint involving offline pre-training for safety, actor-critic architectures for stable learning, and safety-enhancing mechanisms such as action masking and reward shaping [95].

Du et al. introduced the application of Deep Deterministic Policy Gradient algorithms for intelligent multi-zone residential HVAC control, addressing the “curse of dimensionality” that afflicts discrete action spaces in multi-zone scenarios [46]. Their research demonstrated that DDPG-based control could simultaneously achieve 15% cost reduction and 79% improvement in thermal comfort compared to advanced Deep Q-Network approaches, while maintaining excellent generalization capability across different building configurations and pricing scenarios. The success of their approach stemmed from the algorithm's ability to handle continuous action spaces (i.e., temperature setpoints for multiple zones) while learning complex trade-offs between energy costs and occupant comfort through reward function optimization.

Wang et al. advanced this field by integrating load forecasting

directly into the reinforcement learning state space, creating a predictive reinforcement learning framework for ground source heat pump systems [90]. Their innovative approach incorporated future load predictions as state variables alongside current operational parameters, enabling the RL agent to make anticipatory control decisions rather than purely reactive ones. This integration achieved over 10% energy savings compared to conventional control methods, with an additional 0.38% improvement over standard reinforcement learning approaches that did not utilize predictive information.

The emergence of multi-agent reinforcement learning has opened new frontiers for district-level energy management, where multiple buildings coordinate their HVAC operations for collective benefit. Ghofrani et al. developed a sophisticated load synchronization framework that employed neural network-based load forecasting to enable coordinated sinusoidal temperature setpoint scheduling across building communities [34]. Their approach achieved notable results: 12.5% reduction in electricity consumption and 10% reduction in peak demand across a 26-building community, demonstrating the potential of predictive coordination in smart building networks.

Fig. 4 illustrates a closed-loop framework for integrating load forecasting with advanced control strategies. Optimization is depicted as an intrinsic process within the control strategy block rather than a parallel module. Dashed lines indicate feedback pathways for prediction model updating based on operational performance data, forming a complete closed-loop system.

4.3. Emerging learning paradigms: Addressing data scarcity and deployment challenges

While MPC and reinforcement learning represent control strategies that directly govern HVAC system operation, transfer learning, federated learning, and automated machine learning constitute complementary model development paradigms that address practical barriers to deploying advanced prediction models. These learning paradigms are distinct from control strategies in that they primarily concern how prediction models are trained and maintained, rather than how control actions are generated and executed.

The practical deployment of advanced prediction and control systems in real-world buildings has revealed fundamental challenges related to data availability, domain adaptation, and privacy constraints that traditional machine learning approaches struggle to address effectively. Transfer Learning has emerged as a powerful solution for overcoming the cold-start problem in new buildings or retrofitted systems where insufficient historical data precludes effective model training.

Kim et al. developed a comprehensive LSTM-based transfer learning framework that leveraged knowledge from data-rich source buildings to enhance prediction accuracy in data-scarce target buildings [65]. Their methodology employed sophisticated pre-training on standardized building simulation data followed by fine-tuning on limited real-world measurements, achieving substantial performance improvements across diverse building types and climatic conditions. The research demonstrated that transfer learning could enable effective model deployment with minimal on-site data collection, reducing the traditional barrier to advanced building automation adoption.

Federated Learning has gained prominence as a solution to the pervasive data sharing limitations that constrain collaborative model development across building portfolios. Song et al. introduced a multi-stage LSTM federated forecasting framework that enabled multiple buildings to collaboratively train prediction models while preserving data privacy and accommodating heterogeneous data sampling frequencies [88]. Their approach achieved superior performance compared to individual building models while addressing the critical challenge of multi-timescale data integration through innovative interpolation strategies.

Automated Machine Learning represents the most recent paradigm shift toward democratizing advanced prediction capabilities by automating the traditionally expert-intensive processes of model selection, hyperparameter optimization, and feature engineering. Zhang et al. conducted comprehensive evaluations of six open-source AutoML frameworks for building energy prediction, demonstrating that automated approaches could achieve 1.10–18.66% accuracy improvements over traditional manual modeling while requiring minimal domain expertise [59]. Their research revealed that AutoGluon and FLAML frameworks delivered exceptional performance with minimal training time, making advanced AI accessible to building practitioners without extensive machine learning backgrounds.

5. Challenges, opportunities, and future research directions

The remarkable progress achieved in building HVAC load prediction

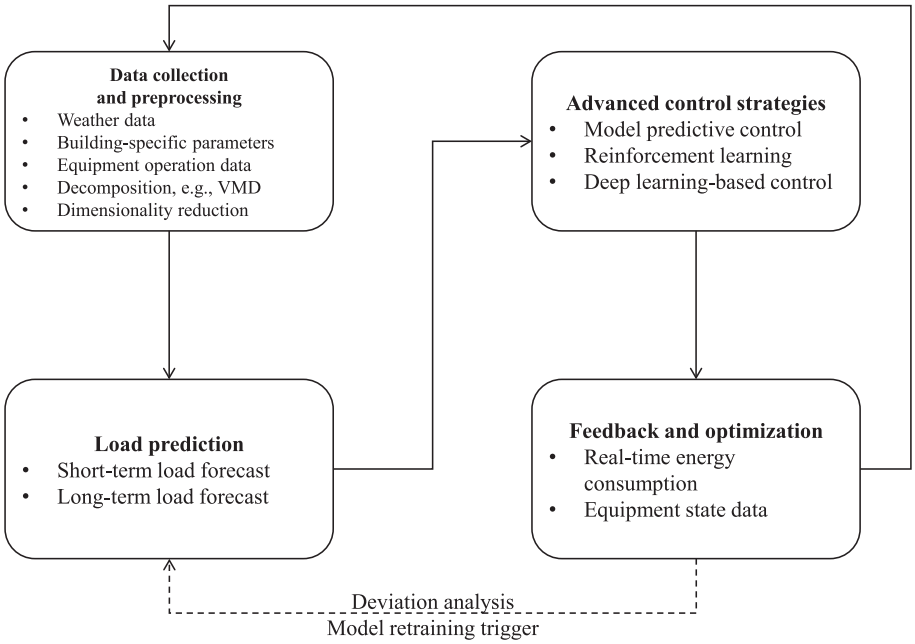


Fig. 4. Framework for integrating load forecasting and advanced control strategies.

methodologies over the past decade has simultaneously illuminated persistent challenges and unveiled transformative opportunities that will shape the future trajectory of this critical research domain. While contemporary approaches have demonstrated impressive accuracy improvements and successful real-world implementations, a candid assessment reveals fundamental limitations that demand innovative solutions and paradigm shifts in how researchers conceptualize, develop, and deploy predictive building energy systems.

5.1. Persistent challenges across multiple dimensions

5.1.1. Data-level impediments and quality assurance complexities

The proliferation of sensor networks and IoT devices in modern buildings has created an apparent data abundance that masks underlying quality and accessibility challenges. Data heterogeneity emerges as a primary concern, where information streams from diverse sources—weather stations, building automation systems, occupancy sensors, and energy meters. These exhibit varying sampling frequencies, measurement accuracies, and temporal alignments [88,96]. Song et al. identified that data sampling inconsistencies across different load types can create synchronization challenges that significantly impact multi-energy prediction accuracy, particularly in integrated energy systems where electrical, thermal, and cooling loads must be forecasted simultaneously [88].

Privacy and data sharing constraints represent significant barriers to collaborative model development, particularly as building owners and facility managers become more protective of operational data that could reveal competitive insights or security vulnerabilities [88,92]. The regulatory landscape surrounding building energy data continues to evolve, with emerging privacy legislation potentially restricting the large-scale data aggregation that has historically enabled breakthrough research achievements [24,36].

Real-time data quality assurance presents ongoing technical challenges that extend beyond simple outlier detection and missing value imputation. Tian et al. revealed that the fundamental challenge of "inaccurate load labels" in training data—where measured cooling capacity oscillates around true load values due to control system dynamics—can degrade prediction model performance by up to 17.55% [63]. This finding highlights the need for sophisticated signal processing techniques that can extract true load signatures from noisy operational measurements.

5.1.2. Model-level limitations and generalization deficits

Despite impressive accuracy achievements in controlled experimental settings, generalization across diverse building types and operational contexts remains problematic for many contemporary prediction models. The extensive analysis by Macas et al. demonstrated that optimal model complexity is strongly coupled with available training data quantity, revealing that complex deep learning models may actually underperform simpler alternatives when deployed in data-constrained environments [9]. This finding challenges the common assumption that more sophisticated algorithms invariably yield superior performance.

Uncertainty quantification and probabilistic forecasting capabilities remain underdeveloped in most current approaches, despite their critical importance for robust control implementation. While some researchers have explored interval prediction through information granulation [5], machine learning-based prediction interval adjustment [37], and Monte Carlo simulation [23,40], systematic approaches to uncertainty quantification that can support risk-informed decision-making in building control systems remain elusive. The research by Wang et al. exposed a fundamental disconnect between prediction accuracy metrics and actual control performance, revealing that models with higher statistical accuracy do not necessarily yield better control outcomes [76].

Several concrete methodological pathways exist for advancing uncertainty quantification in HVAC load prediction. Quantile regression approaches, as demonstrated by Fan et al. [40] through Monte Carlo simulation, can provide prediction intervals alongside point forecasts with relatively modest additional computational overhead. Bayesian deep learning methods, including Monte Carlo Dropout, which can be applied to existing neural network architectures with minimal modification, and Bayesian Neural Networks, offer principled frameworks for estimating epistemic uncertainty arising from limited training data. Conformal prediction, an emerging distribution-free approach, provides mathematically guaranteed prediction intervals without distributional assumptions and has been successfully applied in energy forecasting contexts. Ensemble-based methods, where multiple models with different initializations or architectures provide variance estimates, represent the most immediately practical approach due to their simplicity and compatibility with existing modeling pipelines. For MPC applications specifically, scenario-based optimization using prediction interval bounds can explicitly trade off between energy efficiency and comfort violation risk, as demonstrated by Ma et al. [6]. The research community should prioritize developing uncertainty-aware evaluation metrics that assess both calibration (whether predicted intervals match observed coverage rates) and sharpness (whether intervals are sufficiently narrow to be practically useful).

Interpretability and explainability concerns continue to plague the widespread adoption of sophisticated AI models in building management contexts, where operators require transparent understanding of system behavior for maintenance, troubleshooting, and regulatory compliance purposes [61,71]. The "black-box" nature of deep learning models, while delivering impressive accuracy, creates operational challenges that limit their acceptance in conservative building management environments where system reliability and predictability are paramount.

Practical approaches to model interpretability in HVAC load prediction include post-hoc explanation methods, such as SHAP (SHapley Additive exPlanations) values for quantifying individual feature contributions, and attention weight visualization in Transformer-based architectures, which can reveal which temporal positions and input features the model considers most relevant. Layer-wise Relevance Propagation (LRP) can trace predictions back to input features in deep neural networks. Model-agnostic approaches such as Partial Dependence Plots and LIME (Local Interpretable Model-agnostic Explanations) can provide local explanations for individual predictions. Yang et al. [77] demonstrated the practical value of multi-level decomposition combined with interpretability analysis for air conditioning load forecasting. For building practitioners, the most accessible pathway may be hybrid architectures where an interpretable base model (e.g., linear regression or decision tree) captures the main effects, and a complex model captures the residuals, allowing operators to understand the primary drivers of load predictions while still benefiting from the accuracy of sophisticated methods.

5.1.3. Application-level deployment and integration challenges

The transition from research prototypes to production-ready building energy management systems reveals substantial computational and infrastructure requirements that may exceed the capabilities of typical building automation platforms. The advanced hybrid models demonstrated by researchers such as Zhou et al. and Yang et al., while achieving exceptional accuracy, require computational resources that may not be readily available in standard building management systems [75,86]. For instance, decomposition-integrated prediction pipelines that combine VMD with metaheuristic-optimized deep learning models demand GPU-accelerated computing environments and memory capacities that substantially exceed the specifications of conventional building automation controllers and programmable logic controllers commonly deployed in commercial buildings. The expertise gap between model development complexity and typical building operator capabilities presents a further barrier: the implementation and

maintenance of sophisticated hybrid AI models require specialized knowledge in data science, machine learning, and signal processing that building facility managers rarely possess, creating a dependency on external technical support that increases operational costs and reduces system reliability.

Real-time performance and latency constraints impose additional limitations on model complexity and updating frequency, particularly in applications requiring rapid response to changing conditions or emergency scenarios. Ultra-short-term predictions (sub-hourly) used in real-time MPC implementations must be generated within seconds, yet the sequential execution of signal decomposition, multi-model prediction, and result reconstruction in complex hybrid pipelines can require computation times of minutes or longer on standard hardware. The balance between model sophistication and computational efficiency remains a persistent challenge that researchers must navigate carefully when developing practical building automation solutions. A systematic cost-benefit analysis comparing the marginal accuracy improvements offered by complex hybrid models against the substantially increased computational overhead, maintenance burden, and expertise requirements relative to simpler but more robust alternatives (e.g., well-tuned gradient boosting or single LSTM models) is largely absent from the current literature, yet critically needed to guide practical deployment decisions.

Integration with legacy building systems presents ongoing technical and economic barriers to widespread adoption of advanced prediction and control technologies. Many existing buildings operate with decades-old control infrastructure, including legacy BACnet or Modbus communication protocols, limited-capacity controllers, and proprietary building management systems, that cannot readily accommodate sophisticated AI-based prediction systems, creating a substantial installed base that may remain isolated from contemporary advances unless cost-effective retrofit solutions are developed. The economic viability of retrofitting existing buildings with the sensor infrastructure, computing hardware, and communication networks required to support advanced AI-based prediction remains uncertain for many building types, particularly smaller commercial and residential structures where the absolute energy savings may not justify the upfront investment.

A critical yet often overlooked constraint on practical prediction accuracy is the inherent measurement uncertainty of building sensors. Commercial-grade temperature sensors (±0.2–0.5 °C), humidity sensors (±2–5%), flow meters (±1–3%), and power meters (±1–2%) introduce irreducible noise into both training labels and evaluation metrics [2,42]. This measurement uncertainty establishes a practical accuracy floor that theoretical algorithmic improvements cannot overcome. When prediction models are trained on sensor data containing systematic measurement biases or random noise, the reported prediction errors reflect the model's ability to reproduce sensor readings rather than to predict true physical quantities. Tian et al. demonstrated that the fundamental challenge of "inaccurate load labels", where measured cooling capacity oscillates around true load values due to control system dynamics, can degrade prediction model performance by up to 17.55% [63]. Consequently, the research community must exercise greater rigor in distinguishing between simulation-validated accuracy (where ground truth is precisely known) and field-deployed accuracy (where ground truth is itself uncertain), and future studies should report prediction performance alongside explicit characterization of sensor measurement uncertainty and data quality metrics.

This diagram in Fig. 5 illustrates the key challenges in HVAC load prediction and their corresponding solutions. The primary challenges include data quality issues, limited model generalization, and insufficient uncertainty quantification. To address these, the proposed solutions incorporate physics-informed AI to enhance robustness, integration of LLMs to improve semantic understanding, and federated learning to enable collaborative modeling across buildings while preserving data privacy. These approaches collectively strengthen model performance, adaptability, and uncertainty analysis in complex building environments.

5.2. *Emerging opportunities and future research frontiers*

5.2.1. *Physics-informed artificial intelligence: Bridging first principles and data-driven approaches*

The convergence of physical domain knowledge with artificial intelligence represents one of the most promising frontiers for addressing current limitations in HVAC load prediction. Physics-Informed Neural Networks offer the potential to embed fundamental thermodynamic principles directly into AI model architectures, creating systems that respect physical laws while leveraging data-driven learning capabilities

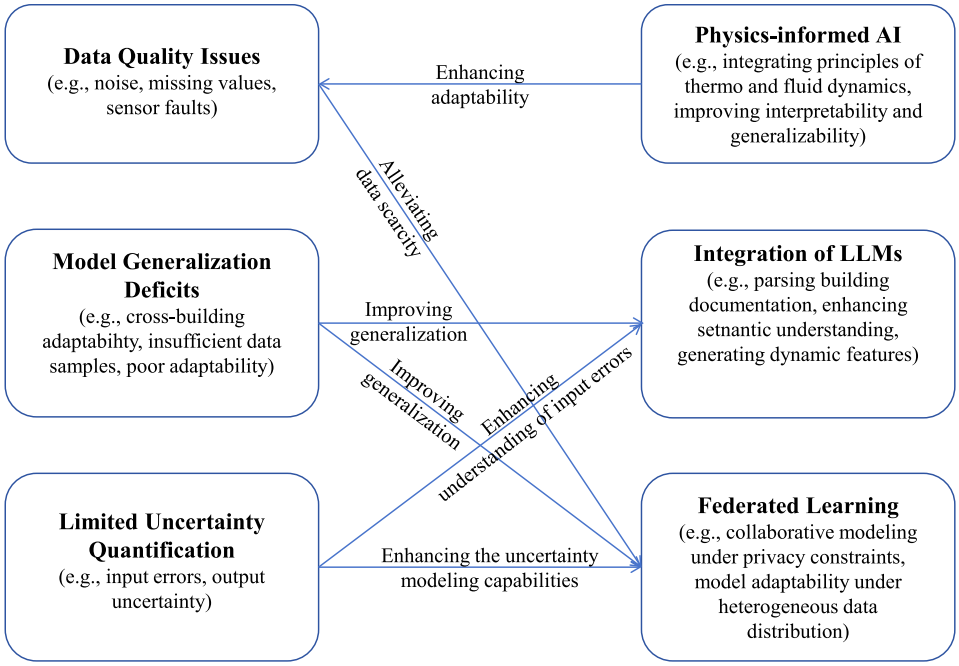


Fig. 5. Conceptual framework for addressing HVAC load prediction challenges: Integrating physics-informed AI, LLMs, and federated learning.

[84,93]. Shi et al. demonstrated preliminary success in this direction by incorporating transient heat transfer theory into deep neural network feature selection, achieving 6.2% prediction accuracy with significantly improved interpretability [84].

It must be noted that physics-informed approaches are in a relatively early stage of application to HVAC load prediction, and their current accuracy (e.g., 6.2% MAPE reported by Shi et al. [84]) does not yet match the best-performing purely data-driven hybrid models. However, the value proposition of PINNs lies not in replacing high-accuracy hybrid models for well-instrumented buildings with abundant data, but rather in addressing the critical challenges of generalization, interpretability, and data efficiency that currently limit the scalable deployment of data-driven methods across diverse building portfolios. The research trajectory suggests that the most promising practical path may be hybrid gray-box frameworks that combine physics-based structural priors with data-driven components, as demonstrated by Lian et al. [93], rather than purely physics-constrained architectures.

Current maturity assessment: PINNs for HVAC load prediction are at an early exploratory stage (Technology Readiness Level 3). While the theoretical framework is well-established in other domains (computational fluid dynamics, structural mechanics), the application to building thermal systems remains nascent. Shi et al. [84] represents one of the few published attempts, achieving 6.2% MAPE, substantially below the performance of mature hybrid models. Key limitations include the difficulty of formulating appropriate physics-based loss terms for complex multi-zone buildings, the computational overhead of physics-constrained optimization, and the limited availability of validated case studies. Initial pilot deployments are likely 3–5 years away, while widespread commercial adoption may require 5–8 years, contingent on the development of standardized physics-informed architectures and validated demonstration projects.

5.2.2. *Large language model integration: Semantic understanding and automated model development*

The rapid development of Large Language Models presents potential opportunities for aspects of building energy management, though their direct application to HVAC load forecasting remains at a very early stage [92,96]. Automated building model generation represents one potentially promising application, where LLMs could potentially interpret architectural drawings, specifications, and operational documentation to automatically construct energy simulation models or configure prediction algorithms without extensive manual input.

Zhang et al. conducted exploratory research demonstrating that LLM-based approaches could generate EnergyPlus simulation models from building design documents in approximately 196 s, opening possibilities for rapid model development and deployment [92]. While such LLM-generated EnergyPlus models do not directly perform HVAC load forecasting, they are relevant to this review in two important ways. First, automatically generated building simulation models can serve as the basis for physics-informed prediction frameworks or as simulation environments for generating synthetic training data when real-world operational data is scarce, a persistent challenge identified in Section 5.1.1. Second, the capability to rapidly create building energy models from design documentation enables faster deployment of simulation-based MPC strategies that rely on building models for load prediction. However, it must be acknowledged that significant validation challenges remain before LLM-generated models can be reliably used for quantitative load forecasting, and this application remains at an early conceptual stage (Technology Readiness Level 2). Future applications may extend to intelligent feature engineering, where LLMs analyze building documentation and operational histories to automatically identify relevant predictive features, and adaptive model selection, where natural language descriptions of building characteristics guide algorithmic choices.

Semantic analysis of building operational data through LLM integration could enable more sophisticated understanding of occupant behavior patterns, equipment performance characteristics, and system anomalies through natural language processing of maintenance logs, occupant feedback, and operational reports. This capability could substantially enhance the contextual understanding that current purely numerical approaches lack.

Current maturity assessment: LLM integration in building energy prediction is at an early conceptual stage (Technology Readiness Level 2). Zhang et al. [92] demonstrated preliminary capability in generating EnergyPlus simulation models from building documents, but this application remains limited to model generation rather than direct load prediction. Significant limitations include the high computational cost of LLM inference, the risk of hallucination in generating building parameters, the lack of domain-specific training data for building energy applications, and the absence of validation frameworks. The most realistic near-term application is likely in automated report generation and natural language interfaces for building operators rather than in direct prediction tasks.

5.2.3. *Multi-agent systems and distributed intelligence*

The evolution toward smart building communities and district energy management necessitates prediction and control frameworks that can operate effectively across multiple buildings while accommodating diverse ownership structures, operational priorities, and privacy requirements [34,88]. Future research in multi-agent systems for building energy management may focus on developing decentralized optimization algorithms that enable buildings to coordinate their energy consumption patterns without requiring centralized control or complete information sharing.

Blockchain-enabled data sharing and secure multi-party computation techniques offer potential pathways for enabling collaborative model training while maintaining data privacy and ownership rights. These technologies could enable large-scale federated learning initiatives that leverage the collective intelligence of entire building portfolios while respecting individual privacy and security requirements.

Current maturity assessment: Federated learning for building energy applications is at an early development stage (Technology Readiness Level 3–4). Song et al. [88] demonstrated feasibility for multi-building collaborative training, but practical challenges remain substantial: communication overhead in building networks with limited bandwidth, handling of non-IID data distributions across buildings with fundamentally different characteristics, convergence guarantees with heterogeneous client data, and the need for standardized federated learning platforms compatible with building automation systems. Pilot deployments in building portfolios under common ownership represent the most feasible near-term pathway.

5.2.4. *Human-centric modeling and behavioral integration*

The incorporation of occupant behavior modeling and personalized comfort preferences represents a critical frontier for enhancing prediction accuracy and control effectiveness in human-occupied buildings [13,26]. Future research may focus on developing adaptive behavioral models that learn individual and collective occupant preferences over time, enabling HVAC systems to anticipate and accommodate human-driven load variations more effectively.

Digital twin integration with real-time occupancy sensing and behavioral analytics could enable unprecedented granularity in load prediction and personalized control strategies. This approach may facilitate the development of human-building interaction models that treat occupants as active participants in building energy systems rather than external disturbances to be managed.

5.2.5. *Lifecycle model management and adaptive learning*

The deployment of AI-based prediction systems in buildings requires sophisticated model lifecycle management capabilities that ensure sustained performance over extended operational periods [54,58]. Future research must address online learning and model adaptation strategies

that enable prediction systems to evolve with changing building characteristics, equipment degradation, and shifting occupancy patterns without requiring complete retraining.

Performance degradation detection and automated model updating represent critical capabilities for ensuring long-term system reliability and accuracy. The development of self-diagnosing prediction systems that can identify when their performance deteriorates and automatically trigger retraining or model replacement procedures will be essential for practical deployment in building management contexts where continuous expert oversight is impractical.

5.3. Future outlook and recommendations

Based on the comprehensive analysis presented in this review, the following recommendations are offered to guide future research and practice:

For researchers: (i) prioritize the development of standardized benchmark datasets and evaluation protocols to enable meaningful cross-study comparisons, analogous to established benchmarks in other AI domains such as ImageNet for computer vision; (ii) advance uncertainty quantification methods, particularly conformal prediction and Bayesian approaches, that can provide calibrated prediction intervals for practical decision-making; (iii) investigate physics-informed architectures that embed thermodynamic constraints into data-driven models to improve generalization and interpretability; (iv) extend validated methodologies to underexplored building types, including industrial facilities, data centers, and mixed-use buildings, which present unique load forecasting challenges [33,74,82]; and (v) develop systematic cost-benefit frameworks that evaluate whether the marginal accuracy improvements offered by complex hybrid models justify increased computational and maintenance overhead.

For practitioners: (i) invest in data quality infrastructure, as preprocessing quality often determines prediction success more than algorithmic sophistication; (ii) adopt hybrid gray-box approaches that balance modeling effort with prediction accuracy for retrofit applications; (iii) consider the complete prediction-to-control pipeline rather than optimizing forecasting accuracy in isolation, as prediction errors may not propagate linearly to control performance [76]; and (iv) evaluate prediction models using downstream control performance metrics rather than relying solely on traditional statistical accuracy measures.

For policymakers: (i) support the development of open building energy datasets and benchmarking platforms to accelerate research progress; (ii) establish standards for prediction model validation and deployment in building energy management systems; (iii) incentivize the integration of advanced forecasting with demand response and grid-interactive building programs; and (iv) address data privacy and sharing regulations that balance innovation needs with building occupant privacy protection.

Fig. 6 illustrates the central role of HVAC load forecasting within the ecosystem of intelligent buildings and energy systems. By integrating with smart grids, distributed energy systems, demand response, thermal energy storage, building automation, and AI-driven decision systems, load forecasting enhances energy efficiency and supports intelligent and carbon-neutral objectives.

6. Conclusions

This comprehensive review has systematically examined the transformative evolution of HVAC load prediction methodologies, revealing a field characterized by significant innovation, increasing sophistication, and growing practical impact across the building energy management landscape. The analysis of 96 peer-reviewed publications spanning the past decade illuminates a clear developmental trajectory from isolated algorithmic improvements toward integrated, intelligence-driven building automation ecosystems that seamlessly unite data science, artificial intelligence, and advanced control theory.

Evidence presented throughout this review demonstrates that the field has progressed far beyond the limitations of traditional statistical approaches and single-algorithm solutions. Contemporary methodologies have embraced hybrid frameworks that systematically combine signal decomposition, metaheuristic optimization, and deep learning architectures to achieve prediction accuracies that were not attainable using conventional approaches. The progression from Mean Absolute Percentage Errors exceeding 15% in early traditional methods to approximately 2–5% under real-world operational conditions (and sub-1% in simulation-based studies) achieved by state-of-the-art hybrid models [1,75,86] represents not merely incremental improvement, but a fundamental leap in predictive capability that enables entirely new

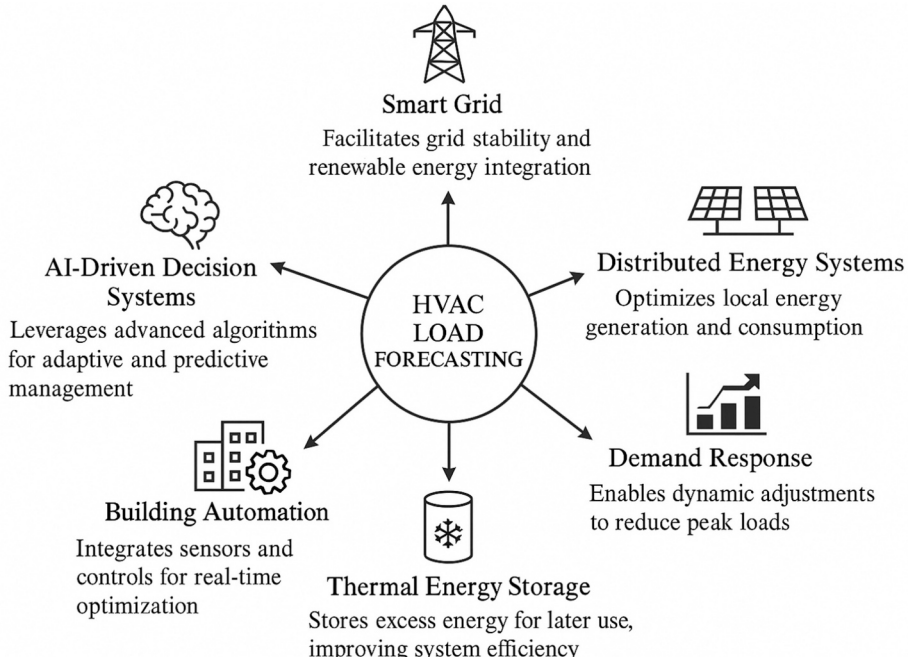


Fig. 6. Schematic diagram of HVAC load forecasting technology integrated with intelligent building and energy systems.

paradigms in building automation.

Prediction-control convergence has emerged as perhaps the most significant development in this domain, transcending the traditional boundaries between forecasting and system operation. The integration of accurate load predictions with Model Predictive Control, reinforcement learning, and multi-agent coordination systems has demonstrated energy savings ranging from 10% to 30% while simultaneously improving occupant comfort and enabling grid-interactive building capabilities [6,31,46,81]. This evolution reflects a mature understanding that prediction accuracy, while important, must ultimately serve the broader objective of optimal building operation rather than existing as an isolated academic exercise.

Data preprocessing and feature engineering methodologies have proven equally transformative, with advanced signal decomposition, intelligent feature selection, and automated data quality assurance procedures often determining model success more significantly than algorithmic sophistication alone [45,53,59]. The systematic application of Variational Mode Decomposition, ensemble-based feature selection, and automated machine learning frameworks has democratized access to high-performance prediction capabilities while reducing the expertise barriers that previously limited widespread adoption.

However, this review has also revealed persistent challenges that demand urgent attention from the research community. The generalization limitations of current models across diverse building types and operational contexts, the inadequate treatment of uncertainty quantification in most approaches, and the ongoing struggle with interpretability in complex AI systems represent fundamental barriers to widespread practical deployment. The absence of standardized benchmark datasets and evaluation protocols represents a significant barrier to meaningful cross-study comparison and objective assessment of methodological progress. Moreover, the emerging challenges related to data privacy, computational requirements, and integration with legacy building systems highlight the need for research approaches that explicitly consider real-world deployment constraints rather than pursuing accuracy improvements in isolation.

Building HVAC load prediction will benefit from emerging paradigms that address these challenges while capitalizing on transformative opportunities. Physics-informed artificial intelligence offers the potential to create models that respect fundamental thermodynamic principles while leveraging data-driven learning capabilities, potentially resolving current generalization and interpretability limitations. The integration of Large Language Models promises to contribute to automated model development and semantic understanding of building operational contexts, while federated learning and multi-agent systems could enable collaborative intelligence across building portfolios without compromising data privacy.

Ultimately, this analysis envisions building energy systems that operate as autonomous, adaptive, and continuously learning entities capable of optimizing their performance across multiple objectives while maintaining seamless integration with broader energy infrastructure. Such systems would combine the precision of physics-based understanding with the adaptability of artificial intelligence, the efficiency of optimized control with the robustness of uncertainty-aware decision-making, and the intelligence of individual building optimization with the collective benefits of community-scale coordination.

This review has several limitations that should be acknowledged. First, the analysis is based on English-language peer-reviewed publications, which may exclude relevant work published in other languages, particularly Chinese-language journals that represent a significant portion of HVAC load forecasting research. Second, the quantitative performance comparisons across studies should be interpreted with caution due to heterogeneous datasets, prediction horizons, building types, climate zones, and validation protocols, a limitation that cannot be fully overcome without community-wide standardized benchmarking. Third, the review focuses primarily on data-driven and hybrid approaches, with limited coverage of purely physics-based simulation methods that continue to play important roles in building energy modeling. Fourth, rapidly emerging technologies such as foundation models and diffusion-based generative models are only briefly discussed, as their application to HVAC load forecasting is too nascent for comprehensive assessment. Finally, the review does not systematically cover all specialized facility types (e.g., data centers, hospitals) that present unique load forecasting challenges distinct from the commercial, residential, and industrial buildings that constitute the primary focus of this work.

As the building sector continues its critical role in global sustainability objectives, the methodologies reviewed in this comprehensive analysis provide the technological foundation necessary to transform buildings from passive energy consumers into active, intelligent participants in the emerging carbon-neutral energy ecosystem. The convergence of data science, artificial intelligence, and control theory in building HVAC load prediction represents not merely a research achievement, but an essential stepping stone toward the intelligent, sustainable built environment that our collective future demands.

CRediT authorship contribution statement

Dapeng Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Yanfeng Gong:** Writing – review & editing, Writing – original draft, Validation, Software, Project administration, Methodology, Investigation, Conceptualization. **Jianan Chen:** Visualization, Validation, Resources. **Rui He:** Resources, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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